THE UNIVERSITY OF DODOMA

COLLEGE OF NATURAL AND MATHEMATICAL SCIENCES

DEPARTMENT OF PHYSICS

COURSE NAME: PHYSICS PRACTICAL III

COURSE CODE: PH 2106

COURSE INSTRUCTOR: MR MPANGO

EXPERIMENT TITLE: COMPARISON OF COOLING RATE OF WATER UNDER FORCED CONVECTION

NATURE OF WORK: INDIVIDUAL WORK

EXPERIMENT NUMBER: 06

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DEGREE PROGRAM: BED-SC-ICT

ACADEMIC YEAR: 2025/2026

EXPERIMENTDATE: 11Th -FEBRUARY-2026

SUBMISSION DATE: 21St-FEBRUARY-2026

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TABLE OF CONTENTS

ABSTRACT 2

THEORY 5

3.1 Heat Transfer During Cooling 5

3.2 Newton's Law of Cooling 6

3.3 Forced Convection and Uniform Temperature 7

3.4 Rate of Cooling and Graphical Interpretation 7

3.5 Validity and Importance of the Theory 8

METHODOLOGY 9

DISCUSSION OF RESULTS 11

CONCLUSION AND RECOMMENDATION 12

REFERENCES 13

ABSTRACT

This experiment was designed to investigate and verify Newton's law of cooling under conditions of forced convection by examining how the temperature of a hot body decreases with time when exposed to a cooler environment. The study specifically focused on establishing the relationship between the rate of fall of temperature and the temperature of the cooling body itself. Forced convection was achieved by continuously stirring the water contained in a calorimeter, ensuring uniform temperature distribution throughout the liquid and enhancing heat transfer between the system and the surrounding air. By carefully recording temperature readings at equal time intervals and analyzing the resulting data graphically, the experiment aimed to demonstrate that the rate of cooling is directly proportional to the temperature excess of the body above its surroundings.

The experimental setup consisted of a calorimeter partially filled with hot water initially heated to approximately 80 °C. A thermometer and stirrer were inserted into the calorimeter immediately to minimize initial heat loss and to allow continuous agitation of the water. The calorimeter was placed on a wooden block to reduce heat transfer by conduction to the bench. Temperature readings were taken at one-minute intervals over a period of twenty minutes using a stopwatch, while the water was continuously stirred to maintain forced convection. The room temperature was measured both before and after the experiment to determine the effective surrounding temperature, which is a critical parameter in the application of Newton's law of cooling. These controlled conditions ensured that convection was the dominant mode of heat transfer and that the assumptions underlying Newton's law of cooling were reasonably satisfied.

The collected temperature–time data revealed a gradual decrease in temperature with time, as expected for a cooling process. A graph of temperature versus time was plotted to visualize the cooling behavior of the system. From this curve, tangents were drawn at several equally spaced temperature points to determine the instantaneous rate of cooling at different temperatures. These gradients represented the rate of fall of temperature at those specific points. The gradients were then plotted against the corresponding temperatures to obtain a second graph, which formed the core analytical tool for testing the validity of Newton's law of cooling under forced convection conditions.

The resulting graph of the rate of cooling against temperature was observed to be approximately linear, indicating a direct proportionality between these two quantities. This linear relationship provides strong experimental support for Newton's law of cooling, which predicts that the rate of cooling is proportional to the temperature difference between the cooling body and its environment. Furthermore, extrapolation of the straight-line graph to the temperature axis produced an intercept close to the measured room temperature. This intercept has a clear physical interpretation: it represents the temperature at which the cooling rate becomes zero, meaning that there is no net heat exchange between the body and its surroundings when both are at the same temperature.

The slope of the gradient–temperature graph was found to be constant, confirming that the proportionality factor, known as the cooling constant, remained approximately unchanged throughout the experiment. This constant depends on factors such as the nature of the calorimeter surface, the surrounding medium, and the efficiency of forced convection produced by stirring.

INTRODUCTION

Heat transfer is a fundamental concept in physics and engineering, governing how energy flows between systems at different temperatures. In everyday life, cooling processes are observed whenever a hot object is left in a cooler environment, such as a cup of hot tea cooling in a room or heated metal losing heat in open air. Understanding the laws that describe such cooling processes is essential in many scientific and technological applications, including thermal engineering, climate control, refrigeration, and material science. One of the most important empirical laws describing cooling phenomena is Newton's law of cooling, which provides a simple yet powerful relationship between the rate of cooling of a body and its temperature relative to its surroundings.

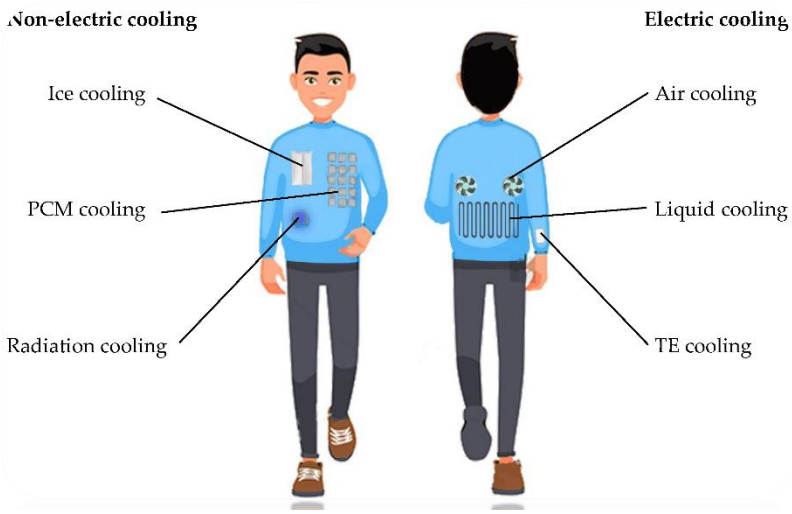

Figure 1: cooling process in human body

Newton's law of cooling states that, for small temperature differences, the rate at which a body loses heat is directly proportional to the difference between the temperature of the body and the temperature of the surrounding environment. Mathematically, this law links the rate of change of temperature of the body to its temperature excess above the surroundings. Although originally formulated as an empirical observation, Newton's law of cooling can be derived from more fundamental principles of heat transfer under specific assumptions. These assumptions include a uniform temperature distribution within the body, constant properties of the surrounding medium, and dominance of convection over other modes of heat transfer.

Heat transfer between a body and its surroundings occurs through three main mechanisms: conduction, convection, and radiation. Conduction involves the transfer of heat through direct molecular interactions within a material or between materials in contact. Convection involves the transfer of heat by the bulk motion of a fluid, such as air or water, surrounding the body. Radiation involves the emission of electromagnetic waves due to the thermal motion of charged particles within the body. In most practical situations, all three modes of heat transfer may occur simultaneously; however, one mechanism often dominates depending on the experimental conditions.

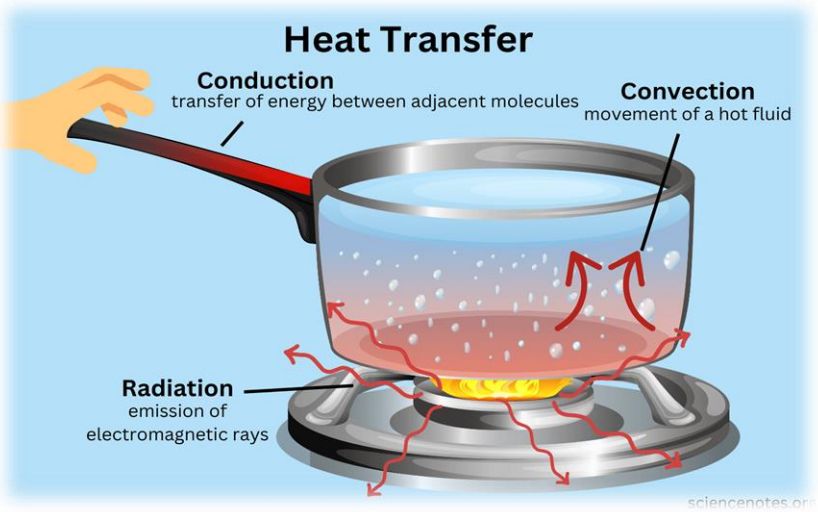

Figure 2: Methods of heat transfer

In the present experiment, convection is the dominant mode of heat transfer. Convection itself can be classified into natural (free) convection and forced convection. Natural convection occurs due to density differences in the fluid caused by temperature gradients, leading to buoyancy-driven fluid motion. Forced convection, on the other hand, occurs when fluid motion is externally induced, for example by stirring, pumping, or the action of a fan. Forced convection generally enhances the rate of heat transfer by increasing fluid mixing and reducing thermal boundary layers around the surface of the cooling body.

Newton's law of cooling is most accurately applicable when the temperature difference between the body and its surroundings is not very large and when the heat transfer coefficient remains approximately constant. Under forced convection conditions, the heat transfer coefficient tends to be more uniform and stable compared to natural convection. This makes forced convection particularly suitable for experimental verification of Newton's law of cooling. By continuously stirring the water in a calorimeter, temperature gradients within the liquid are minimized, ensuring that the measured temperature represents the average temperature of the system at any given time.

THEORY

3.1 Heat Transfer during Cooling

When a body at a higher temperature is placed in surroundings at a lower temperature, heat flows from the body to the surroundings until thermal equilibrium is reached. This heat transfer occurs because of the temperature difference between the body and its environment. The study of how a body cools and how its temperature changes with time is an important part of thermal physics and has many practical applications.

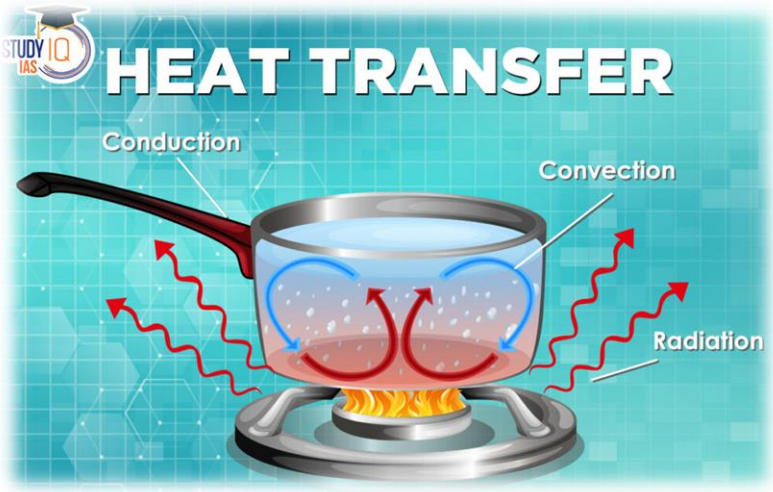

Figure 3: water heat transfer

During cooling, heat is transferred through three main processes: **conduction, convection, and radiation**. In most laboratory cooling experiments, all three processes occur simultaneously; however, one mode usually dominates and determines the overall rate of cooling.

Conduction is the transfer of heat through a material without the bulk movement of the particles. In this experiment, conduction occurs through the walls of the calorimeter and its support. Its effect is reduced by using materials such as wood, which are poor conductors of heat. Radiation involves the emission of heat energy in the form of electromagnetic waves and depends on the temperature and surface properties of the body. Although radiation contributes to heat loss, its effect is relatively small under normal laboratory conditions and is often neglected in theoretical analysis.

Convection is the most significant mode of heat transfer in this experiment. Heat is lost from the surface of the calorimeter to the surrounding air through the motion of air particles. As warm air near the surface rises and cooler air replaces it, heat is continuously carried away, leading to a decrease in temperature of the system.

3.2 Newton's Law of Cooling

Newton's law of cooling states that the rate of loss of heat of a body is directly proportional to the difference between the temperature of the body and the temperature of its surroundings, provided that the temperature difference is small. Expressed mathematically in terms of temperature, the law is given by:

$$dQ/dt = -k(\theta - \theta r)$$

Where
θ is the temperature of the body at time t,
θris the temperature of the surroundings, and
k is a proportionality constant known as the **cooling constant**.

Consider a body of mass m and surface area A at temperature θ, placed in surroundings at constant temperature θr. The body is cooling under forced convection, meaning the surrounding fluid is kept in motion by stirring, ensuring uniform temperature and a constant heat transfer coefficient.

Heat lost by forced convection According to convection, the rate of heat loss is:

$$dQ/dt = hA(\theta - \theta r)$$

Where h is the heat transfer coefficient, A is the surface area, and (θ - θr) is the temperature difference.

Heat lost in terms of temperature change the heat lost causes a decrease in temperature:

$$dQ = mc\, d\theta \;, dQ/dt = -mc\, d\theta/dt$$

(The negative sign indicates loss of heat.)

Equate the two expressions

$$-mc\, d\theta/dt = hA(\theta - \theta r)$$

Rearranging:

$$d\theta/dt = -(hA/mc)(\theta - \theta r)$$

Introduce the cooling constant Let k = had/mc,

$$d\theta/dt = -k(\theta - \theta r)$$

This is Newton's Law of Cooling under forced convection.

The rate of temperature decrease is proportional to the temperature difference.

The cooling constant k depends on surface area, mass, specific heat, and convection efficiency. When θ = θr, dθ/dt = 0, meaning thermal equilibrium is reached.

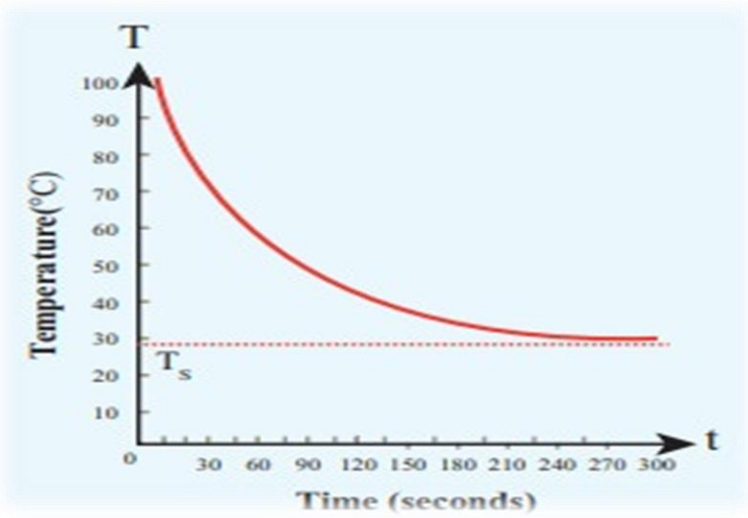

Figure 4: The graph showing cooling relation of temperature against time

Forced convection ensures uniform temperature and a nearly constant heat transfer coefficient, making the law more accurate than under natural convection

The negative sign indicates that the temperature of the body decreases with time. According to this law, the rate of cooling is greatest when the temperature difference is large and decreases as the body approaches the temperature of the surroundings. This explains why a hot object cools rapidly at first and more slowly later.

3.3 Forced Convection and Uniform Temperature

For Newton's law of cooling to be valid, the body must have a uniform temperature throughout. In this experiment, this condition is achieved by continuously stirring the water in the calorimeter. Stirring produces **forced convection**, which ensures that heat is evenly distributed throughout the water, eliminating temperature gradients within the system.

Forced convection also helps to keep the heat transfer coefficient nearly constant during the experiment. As a result, the cooling constant k remains approximately constant, improving the accuracy of the experimental verification of Newton's law of cooling.

3.4 Rate of Cooling and Graphical Interpretation

From Newton's law of cooling, the rate of cooling is defined as:

$$G = -d\theta/dt$$

This shows that the rate of cooling GGG is directly proportional to the temperature excess ($\theta - \theta r$). Therefore, a graph of the rate of cooling versus temperature is expected to be a straight line. The slope of this graph gives the value of the cooling constant k, while the point at which the rate of cooling becomes zero corresponds to the temperature of the surroundings.

The cooling constant depends on factors such as the surface area and material of the calorimeter, the nature of the surrounding medium, and the effectiveness of convection. A larger value of k indicates faster cooling of the system.

3.5 Validity and Importance of the Theory

Newton's law of cooling is valid under the following conditions:

- The temperature difference between the body and surroundings is not very large.
- The temperature of the body is uniform at all times.
- Convection is the dominant mode of heat transfer.
- The heat transfer coefficient remains constant.

METHODOLOGY

For verifying Newton's Law of Cooling under forced convection is a systematic process that ensures accurate observation and analysis of the cooling behavior of a hot body. The experimental approach combines preparation, controlled procedures, precise measurement, and data analysis.

The experiment involves heating water and placing it in a calorimeter to study how its temperature decreases over time when exposed to surroundings at a lower temperature. Forced convection is introduced by continuous stirring, ensuring uniform temperature distribution within the water and maintaining a constant heat transfer coefficient. The methodology emphasizes precise temperature measurements at regular intervals, recording of surrounding conditions, and analysis of the data through graphical methods to determine the rate of cooling and verify the proportionality predicted by Newton's Law

1. **Preparation of Materials**: Collect a calorimeter, hot water (around 80°C), thermometer, stirrer, stopwatch, wooden block, and supporting beakers. Place the calorimeter on a wooden block to minimize conductive heat loss.
2. **Filling the Calorimeter**: Heat water to the desired temperature and fill the calorimeter to about half its capacity. Insert the thermometer and stirrer immediately to prevent initial heat loss and start measurements promptly.
3. **Forced Convection**: Stir the water continuously throughout the experiment to maintain uniform temperature distribution. Forced convection ensures the heat transfer coefficient remains nearly constant and the assumptions of Newton's Law of Cooling are satisfied.
4. **Temperature Recording**: Use a stopwatch to record the temperature of the water every one minute for a total of 20 minutes. Record the surrounding room temperature before and after the experiment to determine the environment temperature accurately.
5. **Procedural Steps**:

    I. Place the calorimeter on the wooden block.
    II. Fill with hot water to half capacity.
    III. Insert thermometer and stirrer.
    IV. Start continuous stirring to achieve forced convection.
    V. Start the stopwatch and record temperatures at one-minute intervals.
    VI. Continue recording for 20 minutes.
    VII. Note the final room temperature.

Safety Precautions

Handle hot water carefully to prevent burns. Read the thermometer at eye level to avoid parallax error. Maintain consistent stirring to ensure uniformity.

This methodology provides a structured and controlled approach to experimentally verify Newton's Law of Cooling, ensuring accurate and reliable results under forced convection conditions.

DATA AND DATA INTERPRETATION

The cooling process of water in the calorimeter was observed and recorded over a period of 20 minutes under forced convection. A graph of temperature against time illustrates the gradual decrease in temperature, which indicates the typical behavior of a cooling process. Tangents were drawn at selected points on this temperature–time graph to determine the instantaneous rate of cooling (G = -dθ/dt). These rates were then plotted against the corresponding temperatures to verify the linear relationship predicted by Newton's Law of Cooling.

From the graph of G against θ the slope

$$slope = -\frac{\Delta G}{\Delta \theta}$$

$$Slope = -\frac{3.8 - 2}{70 - 50} = \frac{1.8}{20}$$

$$The\ slope\ of\ the\ graph\ is - 0.087 min^{-1}$$

The value of slope represents cooling constant (k) of the system, which means that for every $1^{\circ}c$ increase in temperature of the body above the surrounding, the rate of cooling increases by approximately $0.087 min^{-1}$under forced convection

The temperature versus time graph shows a steady decrease in water temperature over the observation period, confirming that the body is losing heat to its surroundings. The instantaneous gradients obtained from the tangents represent the rate of cooling at various temperatures. When these rates are plotted against the corresponding temperatures, the resulting graph is approximately linear, demonstrating that the rate of cooling is directly proportional to the temperature excess of the body above the surroundings.

The intercept on the temperature axis of the gradient graph corresponds closely to the room temperature, which represents the point at which the rate of cooling becomes zero and the body reaches thermal equilibrium with its environment. The slope of the G versus θ graph provides the cooling constant (k), which reflects factors such as the effectiveness of forced convection, the surface characteristics of the calorimeter, and the properties of the water.

Minor deviations from perfect linearity in the graph can be attributed to experimental errors such as heat loss by radiation, non-uniform stirring, or slight fluctuations in room temperature. Despite these small deviations, the observed cooling trends and analysis of the rates support Newton's Law of Cooling, confirming a clear linear relationship between the rate of temperature decrease and the temperature difference from the surroundings.

DISCUSSION OF RESULTS

The experiment successfully demonstrated the cooling of water under forced convection and verified Newton's Law of Cooling. The observed decrease in temperature over time followed the expected trend, with the water cooling rapidly at the beginning and more slowly as it approached the surrounding room temperature. This behavior aligns with the theoretical prediction that the rate of cooling is proportional to the temperature difference between the body and its surroundings.

The graph of the rate of cooling against temperature showed an approximately linear relationship, indicating that the assumptions of Newton's Law of Cooling were reasonably satisfied. The slope of the graph, representing the cooling constant (k), remained fairly constant throughout the experiment. This confirms that forced convection effectively maintained uniform temperature within the water and ensured a nearly constant heat transfer coefficient. The intercept of the graph on the temperature axis corresponded closely to the room temperature, indicating that thermal equilibrium was correctly identified.

Minor deviations from perfect linearity were observed and can be attributed to practical experimental limitations. These include slight variations in room temperature, minor heat loss through radiation, and small inconsistencies in stirring. Such factors are common in laboratory experiments but do not significantly affect the overall validity of the results. The experimental findings strongly support the theoretical expectations, showing that the rate of fall of temperature is directly proportional to the temperature excess above the surroundings.

Overall, the discussion of results emphasizes the importance of controlled experimental conditions, accurate measurement, and careful data analysis. The experiment not only verified Newton's Law of Cooling under forced convection but also reinforced essential laboratory skills, including graphical analysis, interpretation of gradients, and consideration of sources of error.

CONCLUSION AND RECOMMENDATION

The experiment successfully verified Newton's Law of Cooling under forced convection. The temperature of the water decreased over time as expected, and the rate of cooling was found to be directly proportional to the temperature excess above the surroundings. The graph of the rate of cooling against temperature was approximately linear, with the slope representing the cooling constant (k) and the intercept corresponding to the room temperature. Minor deviations were observed due to experimental limitations, but the overall results were in good agreement with theoretical predictions. The study also highlighted the importance of forced convection in maintaining uniform temperature and ensuring accurate verification of the law

For future experiments, it is recommended to minimize external sources of error by using a calorimeter with better insulation to reduce heat loss by radiation. Temperature readings could be recorded using digital thermometers to improve accuracy and reduce human error. Additionally, maintaining a strictly controlled room temperature and ensuring more uniform stirring would enhance the precision of the results. Extending the experiment to different liquids or varying the surface area of the calorimeter could provide further insights into the effects of material properties and geometry on the cooling constant.

REFERENCES

Serway, R. A., & Jewett, J. W. (2018). *Physics for Scientists and Engineers* (10th Ed.). Cengage Learning.

Halliday, D., Resnick, R., & Walker, J. (2020). *Fundamentals of Physics* (11th Ed.). Wiley.

University Practical Physics Notes. (Nod). Heat and Thermodynamics. Retrieved from *University Physics of Dodoma Physics Lab Manual* 2023/2024.

Tipler, P. A., & Mosca, G. (2007). *Physics for Scientists and Engineers* (6th Ed.). W. H. Freeman.

Cengel, Y. A., & Ghajar, A. J. (2015). *Heat and Mass Transfer: Fundamentals and Applications* (5th Ed.). McGraw-Hill Education.